One Flow, Two Notions of Quality: Sampling Budgets for Image Inpainting

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Abstract

Increasing inference computation may improve image inpainting, but its benefits depend on how reconstruction quality is measured. We study this relationship using conditional flow matching (FM), varying function evaluations within each trained model. On 64 × 64 Oxford-IIIT Pet images, we compare FM with pixel-loss and perceptually supervised regression under matched CNN architectures, training data, and update budgets. Across two mask geometries and two training seeds, development PSNR is highest at two evaluations, whereas LPIPS is lowest at eight. Larger tested budgets worsen both metrics. The development-selected eight-evaluation model improves mean PSNR and LPIPS over both regression baselines on held-out test images. Feature supervision narrows the perceptual gap, but FM retains an 11.8% reduction in mean LPIPS and a 0.437 dB gain in mean PSNR over perceptual regression, at approximately 7.4 times its single-image inference latency. These results show that more sampling does not consistently improve reconstruction, and that inference budgets should reflect the target quality criterion and computational cost.

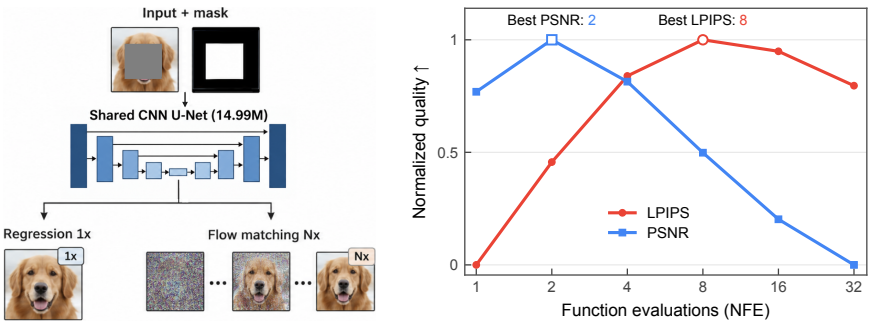


Figure 1: **Different quality criteria favor different sampling budgets.** Left: schematic comparison of independently trained models with a common architecture, updating only missing pixels. Right: seed- and geometry-averaged development scores, min–max normalized per metric with LPIPS reversed (higher is better).

1 Introduction

Image inpainting reconstructs missing image content from the surrounding visible context. Reconstruction quality can be assessed from different perspectives. Pixel fidelity measures how closely the predicted pixels match the reference image, while perceptual similarity measures agreement in a learned feature space [Zhang et al., 2018]. These criteria need not favor the same completion. Inpainting also involves a computational choice: a model can predict the missing content in a single forward pass or generate it through multiple sampling steps. Understanding what additional inference computation contributes therefore requires examining both the quality criterion and the cost of obtaining a completion.

Preprint. **Code:** github.com/Yongji0427/flow-inpainting-study

Flow matching learns a velocity field that generates samples through numerical integration [Lipman et al., 2023]. For image inpainting, conditioning this field on the visible image and mask allows it to generate the missing content while preserving the observed region. The number of function evaluations (NFE) provides a controllable inference budget within a trained model. This setting raises two related questions: whether increasing the sampling budget consistently improves reconstruction quality, and whether pixel fidelity and perceptual similarity favor the same budget. Studying these questions within a fixed model separates changes in sampling computation from changes in architecture or training.

We investigate these questions on 64×64 Oxford-IIIT Pet images using two mask geometries and two training seeds. We sweep the sampling budget within each trained conditional FM model and compare its performance with independently trained regression models using the same 14.99M-parameter CNN architecture, training data, and update budget. Pixel-loss regression provides a single-pass reconstruction baseline. Since feature-based supervision can improve feedforward image transformations [Johnson et al., 2016], we also include perceptual regression to examine whether this supervision narrows the quality gap with FM. Together, these comparisons characterize the effect of sampling within FM and how its quality and computational cost compare with single-pass alternatives.

The development experiments reveal different preferred sampling budgets for the two quality criteria. Among the tested settings, two evaluations achieve the highest PSNR, whereas eight achieve the lowest LPIPS. Increasing the budget beyond eight evaluations worsens both metrics. On held-out test images, the development-selected eight-evaluation configuration achieves higher mean PSNR and lower mean LPIPS than both regression baselines, including the perceptually supervised model. These improvements require approximately 7.4 times the single-image inference latency of perceptual regression. The results show that sampling budget is a quality-dependent design choice: additional computation can improve inpainting, but its value depends on which aspect of reconstruction quality is prioritized and the associated inference cost.

2 Related Work

Flow models and inference budgets. Flow Matching learns a generative velocity field along prescribed conditional probability paths, with samples obtained through numerical integration [Lipman et al., 2023]. This formulation makes the number of function evaluations a controllable inference cost. MeanFlow instead learns interval-average velocity for one-step generation [Geng et al., 2025], illustrating a different route to reducing sampling computation through the learned prediction target. Its one-step predictor therefore differs from applying a single Euler step to a standard FM model.

In image restoration, Restora-Flow adapts an existing generative flow prior through degradation-mask guidance and trajectory corrections, without task-specific retraining [Hadzic et al., 2026]. Its super-resolution experiments examine integration steps, correction steps, and the relationship between quality and runtime. Agarwal et al. [2025] compare diffusion, conditional FM, and MeanFlow using a common TinyUNet on CIFAR-10, and extend FM to mask-guided inpainting through task-specific fine-tuning. These studies connect flow-based generation to restoration and computational efficiency. Our study examines how sampling budgets affect pixel fidelity and perceptual similarity within each frozen inpainting model, alongside regression comparisons matched in architecture and training updates.

Architectures and supervision for restoration. Restoration networks must combine spatial detail with information from the surrounding context. The U-Net architecture combines a contracting path with an expanding path and skip connections, providing a foundation for dense image prediction [Ronneberger et al., 2015]. LaMa addresses large missing regions using Fourier convolutions with an image-wide receptive field, a high-receptive-field perceptual loss, and large training masks [Suvorov et al., 2022]. These approaches highlight architecture, spatial context, and supervision as distinct influences on reconstruction quality.

Comparisons between generative and regression models must account for these influences. Palette studies conditional diffusion across image-to-image tasks, including inpainting, and compares diffusion with regression under an identical neural architecture [Saharia et al., 2022]. This provides a precedent for controlling architecture when comparing different prediction formulations. Supervision also matters within feedforward models. Johnson et al. [2016] demonstrate the value of pretrained

feature losses for style transfer and super-resolution. Their results motivate considering perceptually supervised regression alongside pixel-loss regression when assessing the benefits of iterative generation.

Reconstruction quality and evaluation. Different notions of image quality require different interpretations. Blau and Michaeli [2018] formalize a tradeoff between reconstruction distortion and agreement with the natural-image distribution. LPIPS measures a different quantity: similarity to a paired reference in deep feature space, evaluated against human perceptual judgments [Zhang et al., 2018]. A comparison of PSNR and LPIPS therefore examines two reference-based criteria rather than directly measuring the distributional perception–distortion tradeoff. Cohen et al. [2024] further distinguish posterior sampling from producing semantically meaningful alternatives, showing that random sampling need not reveal a useful range of plausible restorations. Together, these distinctions separate reference fidelity, distributional realism, and semantic diversity. We focus on the first of these, using pixel-level and feature-level comparisons to characterize reconstruction quality across sampling budgets.

3 Data

We use the Oxford-IIIT Pet dataset [Parkhi et al., 2012], which contains 37 pet categories. The original RGB images serve as reconstruction targets, while the masked RGB images and binary masks provide the conditioning inputs. The dataset provides foreground trimaps and, for a subset of images, head-region annotations. We use these annotations solely to construct comparable occlusion masks rather than for dataset organization.

The public dataset provides 3,680 training and validation images alongside 3,669 test images. An inherited manifest from a prior classification project removes 26 near-duplicate images and divides the remainder into 2,924 training and 730 validation images. We sample a fixed random subset of 128 validation images to form our development pool. After applying mask-matching criteria, we retain 2,812 training images and 124 development images (Table 1). The official test split is held out from model training and configuration selection. During test mask matching, 84 images fail the spatial constraints, leaving 3,585 test images for the final evaluation. Head-box annotations are unavailable for the test files, so mask matching on the test split relies entirely on trimaps and spatial constraints.

Table 1: Dataset sizes before and after mask matching. Each pair contains one large and one fragmented mask.

Partition	Offered	Retained	Pairs/image	Role
Training	2,924	2,812	16	Parameter optimization
Development	128	124	4	Analysis and configuration choices
Official test	3,669	3,585	4	Fixed-setting inpainting evaluation

All images undergo EXIF orientation correction, RGB conversion, bicubic center-fit resizing to 64×64 pixels, and normalization to $[-1, 1]$. During training, images and masks are jointly flipped horizontally with probability 0.5. For development and testing, we use fixed masks and a pre-generated noise bank to ensure identical evaluation conditions across all models and sampling budgets.

The paired masks evaluate different spatial contexts while removing exactly 1,638 of 4,096 pixels, approximately 40% of the image. A large mask removes a randomly positioned and oriented 31×53 rectangle with five corner pixels removed. A corresponding fragmented mask introduces one-pixel-wide visible slits and compensates for the removed area within the same quadrants. Pair construction preserves missing area, quadrant counts, and central 16×16 coverage. Annotation tolerances approximately match foreground and head-box occlusion across the training and development pairs.

4 Approach

Our controlled comparison uses a common restoration backbone to isolate differences in the learning objective and inference procedure. Let $x \in [-1, 1]^{3\times H\times W}$ denote an original image, $m \in \{0, 1\}^{H\times W}$ the missing-region mask, and $y = (1 - m) \odot x$ the visible observation. Mask

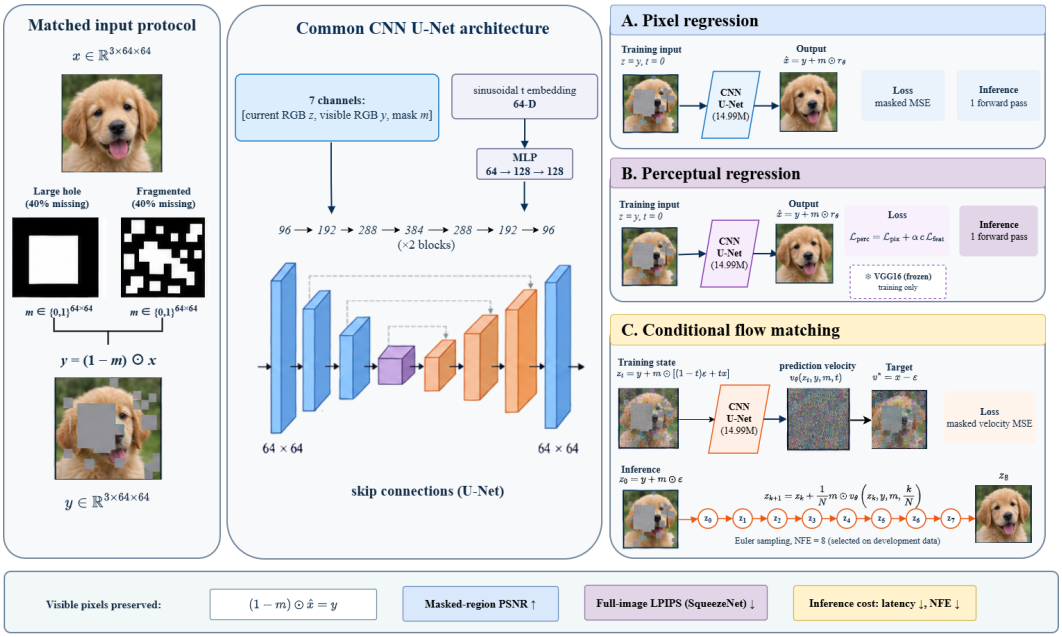


Figure 2: **Overview of the controlled comparison.** Three models use the same U-Net architecture and are trained independently. Pixel and perceptual regression use one forward pass, while conditional FM uses Euler sampling (NFE $= 8$ shown). VGG16 supplies training supervision for perceptual regression.

multiplication broadcasts over the color channels. Each model exactly preserves y outside the mask. Pixel regression, perceptual regression, and conditional FM are trained independently from scratch.

4.1 Common CNN Architecture and Regression Objectives

The shared backbone is a residual U-Net [Ronneberger et al., 2015] containing 14,986,435 trainable parameters. It features channel widths of $(96, 192, 288, 384)$, three downsampling stages, two bottleneck blocks, and nearest-neighbor upsampling with concatenated skip connections. It uses GroupNorm and SiLU activations. A 128-dimensional MLP supplies a sinusoidal time embedding. The network accepts seven input channels: a current RGB state, the visible RGB image y, and the mask m.

Regression models evaluate the network in a single forward pass, setting the current RGB state to y and the time embedding to zero. The network predicts the missing pixels $r_\theta(y, m)$, producing the completion $y + m \odot r_\theta(y, m)$. The pixel-regression model minimizes mean squared error over the missing pixels:

$$\mathcal{L}_{\text{pix}} = \mathbb{E}_{x,m}\left[\frac{\|m \odot (r_\theta(y, m) - x)\|_2^2}{3\sum_p m_p}\right]. \tag{1}$$

To isolate the effect of perceptual supervision, we include a perceptual-regression control. We add a feature loss in this model computed by a frozen ImageNet VGG16 network [Johnson et al., 2016] at layer indices 3, 8, and 15. The combined objective is:

$$\mathcal{L}_{\text{perc}} = \mathcal{L}_{\text{pix}} + \alpha c\, \mathcal{L}_{\text{feat}}. \tag{2}$$

The scale $c = 0.33623046$ matches initial pixel and feature loss magnitudes. On seed 0, we compared $\alpha \in \{0.1, 1.0\}$ using development LPIPS. Increasing α lowered LPIPS from 0.11735 to 0.11490 while reducing PSNR from 16.042 to 15.867 dB, leading us to select $\alpha = 1.0$ for repetition on seed 1. VGG16 provides training supervision, while SqueezeNet is used for LPIPS evaluation.

4.2 Conditional Flow Matching and Sampling

The conditional FM model learns a velocity field that interpolates between Gaussian noise and the clean missing region [Lipman et al., 2023]. For independent $\epsilon \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, the training state and objective are:

$$z_t = y + m \odot \big((1-t)\epsilon + tx\big), \tag{3}$$

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{x,m,t,\epsilon}\left[\frac{\|m \odot (v_\theta(z_t, y, m, t) - (x - \epsilon))\|_2^2}{3\sum_p m_p}\right]. \tag{4}$$

The network predicts a time-dependent velocity from the current state and the visible observation. At inference, sampling begins at $z_0 = y + m \odot \epsilon$ and proceeds via uniform forward Euler integration. For a budget of N function evaluations (NFE), each update is:

$$z_{k+1} = z_k + \frac{1}{N} m \odot v_\theta(z_k, y, m, k/N), \qquad k = 0, \ldots, N-1. \tag{5}$$

We mask the velocity at every step to preserve visible pixels. The final image is clipped to $[-1, 1]$. In our experiments, we sweep $N \in \{1, 2, 4, 8, 16, 32\}$ on development images, reusing the identical initial noise bank at each budget.

4.3 Training and configuration control

All trainable CNNs start from scratch. Within each seed, methods share initial weights, image draws, mask-pair draws, and horizontal flips. Each batch contains 64 images with balanced mask geometries, sampled with replacement from the training images and their 16 available mask pairs. Both seeds use 30,000 AdamW updates, weight decay 10^{-4}, gradient-norm clipping at 1.0, and an exponential moving average (EMA) with decay 0.995. Final comparisons use the EMA checkpoint at the pre-specified 30k endpoint.

The shared learning-rate schedule multiplies a 100-update linear warm-up by a cosine decay from 2×10^{-4} to 4×10^{-5} over the first 2,000 updates. The rate stays constant through 10,000, rises linearly to 8×10^{-5} at 11,000, and decays by cosine to 10^{-5} at 30,000. We retain this inherited schedule across methods and train with BF16 autocasting on the recorded RTX 4090 system. The comparison matches update counts, while perceptual regression incurs additional training computation through its frozen VGG feature pass. Section F of the supplementary material provides software versions and full settings.

5 Experiments

5.1 Evaluation Protocol

We measure pixel fidelity using PSNR on the missing region. To measure perceptual similarity, we use the calibrated SqueezeNet LPIPS model of Zhang et al. [2018], evaluated at native 64×64 resolution on the full composed image. For masked PSNR, we map the clipped completion $\hat{x}$ and reference x from $[-1, 1]$ to $[0, 1]$, defining $\hat{x}_{01} = (\hat{x} + 1)/2$ and $x_{01} = (x + 1)/2$. We then compute:

$$\mathrm{MSE}_m = \frac{\|m \odot (\hat{x}_{01} - x_{01})\|_2^2}{3\sum_p m_p}, \qquad \mathrm{PSNR}_m = -10\log_{10}\big(\max(\mathrm{MSE}_m, 10^{-12})\big). \tag{6}$$

We first characterize the sampling profile on development images and then evaluate the fixed conditional FM configuration on the test split. Test comparisons evaluate the final 30,000-update EMA checkpoints. Every image uses four fixed mask pairs. To isolate algorithmic differences, we use four fixed Gaussian initializations per FM condition and score each completion individually before averaging the results. Test differences use 10,000 bootstrap resamples to describe image-sampling uncertainty conditional on the two trained models.

5.2 The Effect of Inference Computation

Additional sampling changes the quality profile in three stages (Figure 3). The first stage improves both metrics simultaneously. Averaged over seeds and geometries, one evaluation yields an LPIPS of 0.11616 and a PSNR of 16.432 dB, while two evaluations improve these metrics to 0.10929 and 16.530 dB. The second stage favors perceptual similarity over pixel fidelity. At eight evaluations, LPIPS reaches its lowest value of 0.10111, but PSNR decreases to 16.317 dB. The third stage reduces both aspects of reconstruction quality, with 32 evaluations worsening LPIPS to 0.10418 and PSNR to 16.105 dB.

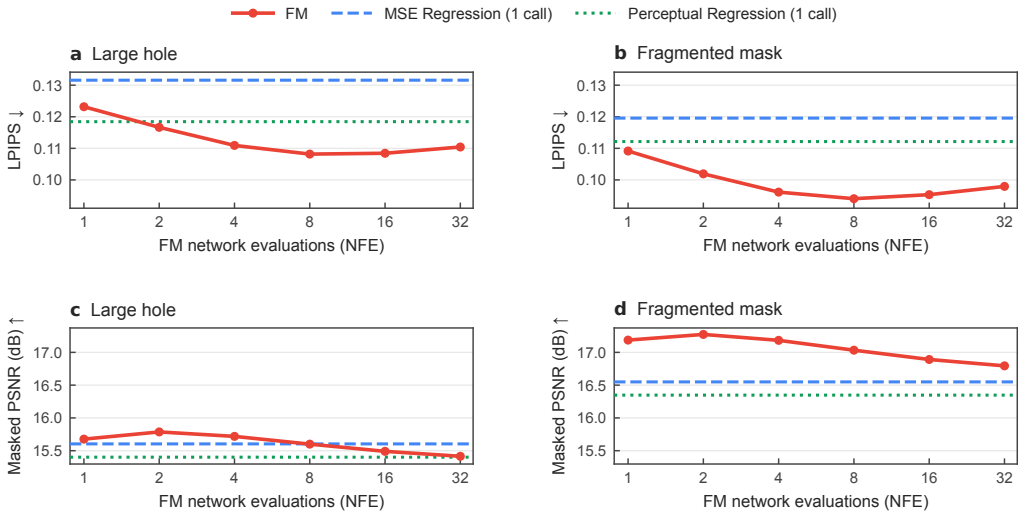


Figure 3: **Pixel fidelity and perceptual similarity favor different sampling budgets.** Frozen FM models use the same development images, four mask pairs per image, and four noise samples across NFE settings. Curves show means over two training seeds; horizontal lines show the mean regression baselines.

These patterns hold across both training seeds and mask geometries. The large-mask curves exhibit higher LPIPS and lower PSNR than the fragmented-mask curves, demonstrating that geometry alters baseline difficulty without changing the optimal sampling pattern. We select the configuration with eight evaluations (FM8) for subsequent testing because it achieves the best perceptual similarity.

5.3 Test Evaluation and Perceptual Control

The fixed FM8 configuration improves both overall metrics on the 3,585 test images (Table 2). Relative to pixel regression, LPIPS falls from 0.13119 to 0.10621, a 19.0% reduction, and PSNR increases by 0.254 dB. The pixel-fidelity improvement is concentrated in fragmented masks, where the gain is 0.479 dB, compared with 0.029 dB for large masks. Furthermore, the large-mask PSNR interval for seed 0 includes zero. The advantage of FM is therefore clearest in perceptual similarity across both geometries and in pixel fidelity strictly for fragmented masks.

Table 2: Mean test quality over two training seeds. LPIPS is lower-is-better and masked PSNR (dB) is higher-is-better. Overall values weight both geometries equally. † denotes perceptual regression with development-selected $\alpha = 1$.

		Large		Fragmented		Overall	
Method	NFE	LPIPS	PSNR	LPIPS	PSNR	LPIPS	PSNR
Pixel regression	1	0.13733	15.461	0.12505	16.350	0.13119	15.906
Perceptual regression†	1	0.12429	15.287	0.11663	16.158	0.12046	15.723
Conditional FM	8	**0.11348**	**15.490**	**0.09895**	**16.829**	**0.10621**	**16.159**

The pixel-regression baseline optimizes reconstruction error alone. We therefore include a perceptual-regression control to examine whether feature supervision narrows the quality gap with FM. Its loss weight is selected using development LPIPS. As expected, strengthening the regression objective

reproduces the tradeoff observed during development, improving LPIPS over pixel regression while reducing PSNR.

FM8 retains an overall advantage in both mean metrics over the perceptual regressor. Its LPIPS is lower by 0.01425 (an 11.8% relative reduction), and its PSNR is higher by 0.437 dB. Both directions hold for the image-averaged results within each seed and geometry. FM8 thus retains higher mean reconstruction quality than the tested perceptual-regression baseline.

5.4 Inference Cost

This quality advantage comes with a substantially higher inference cost. Measured on the same GPU, the median single-image latency is 28.41 ms for FM8 and 3.82 ms for perceptual regression (Table 3). This represents approximately a 7.4-fold increase. Perceptual and pixel regression have similar inference times because the VGG feature loss is evaluated only during training. Consequently, users must weigh the 11.8% perceptual improvement against the 7.4-fold latency penalty.

Table 3: GPU inference time for one output per image. Batch-1 columns report median and 95th-percentile latency. Batch-64 reports amortized time per image (median batch time divided by 64).

Method	NFE	Median (ms)	P95 (ms)	Batch-64 (ms/image)
Pixel regression	1	3.77	3.89	0.298
Perceptual regression	1	3.82	3.94	0.298
Conditional FM	8	28.41	28.71	2.336

5.5 Training Dynamics and Generalization

To understand the learning behavior behind the chosen checkpoints, we evaluate fixed probes on training and development images across 10,000, 20,000, and 30,000 updates (Figure 4). Both flow and regression methods exhibit decreasing training-probe objectives and rising development-probe objectives, which typically indicates overfitting.

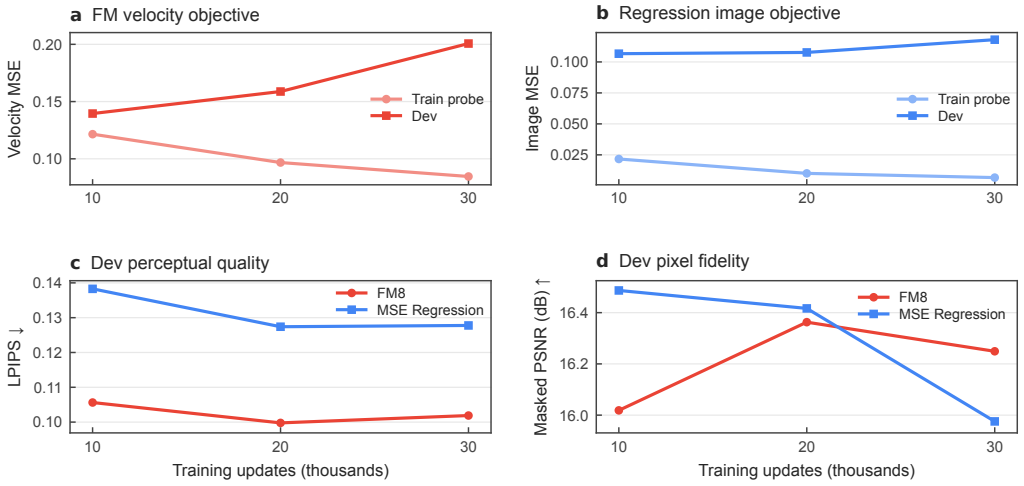


Figure 4: **Training objectives and reconstruction metrics follow different trajectories.** Top: EMA objectives on fixed training and development probes. Bottom: FM8 and pixel-regression metrics using the first mask pair per image and equal geometry weighting. All curves show means over two training seeds.

However, task metrics follow a different trajectory. FM reconstruction improves from 10,000 to 20,000 updates despite the rising development velocity loss, showing that the optimized objective alone is insufficient to track task quality. Later checkpoints reveal a limit to continued training. For fragmented masks, mean FM LPIPS increases from 0.09106 at 20,000 updates to 0.09523 at 30,000 updates, while PSNR falls from 17.153 to 16.941 dB. The final comparison uses the pre-specified 30,000-update budget for all models. Intermediate checkpoints characterize the learning trajectory.

5.6 Qualitative Behavior and Limitations

Figure 5 contrasts a favorable completion with a structural failure under both mask geometries. Among three archived test images, the favorable image minimizes mean FM LPIPS across geometries, while the failure image maximizes FM LPIPS within each geometry. In the favorable example, the missing region covers a dark background and part of the animal's body boundary. Pixel regression produces a lighter, blurred patch, perceptual regression reduces these artifacts, and FM8 more closely recovers the dark background.

In the failure example, the mask covers much of the animal's face and upper body. All three methods fail to recover the correct facial structure, and the visible slits in the fragmented mask do not resolve this failure. FM8 produces incorrect dark and colored structures in the large hole despite achieving lower LPIPS than either regressor for this condition. For the fragmented mask, perceptual regression has lower LPIPS than FM8 (0.11198 versus 0.12094). The FM condition-level scores average four fixed noise samples, while the figure displays the first sample. These examples show that lower LPIPS can coexist with incorrect facial structure.

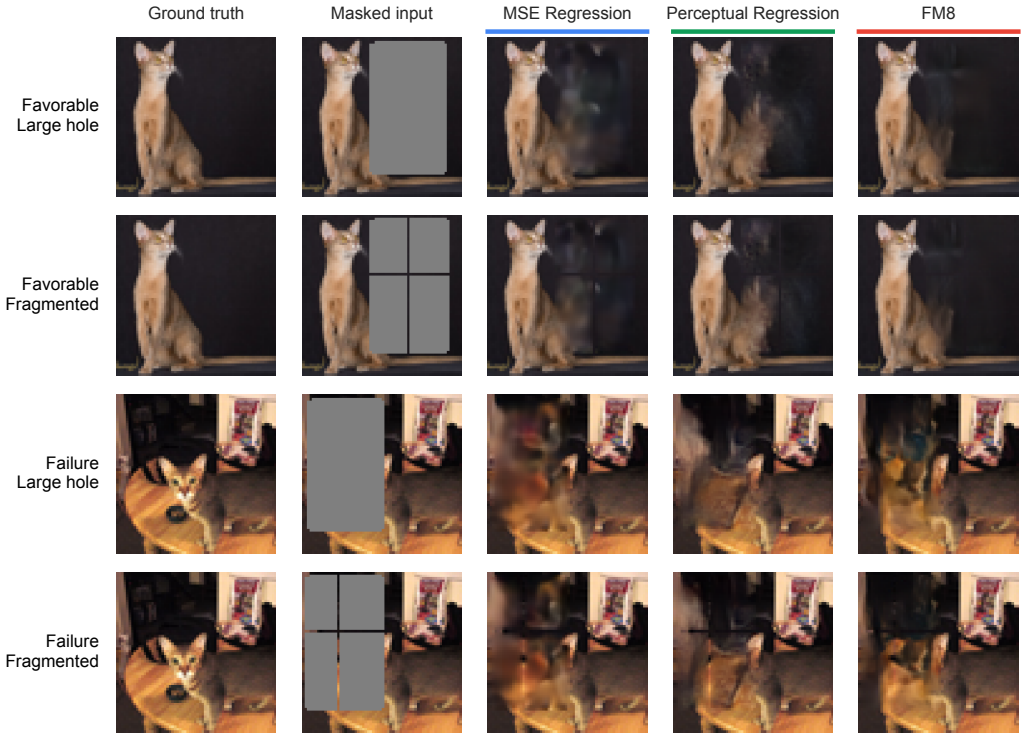


Figure 5: **Favorable and failed test completions selected from three archived images.** Two images are shown under both mask geometries, using seed 0, 30k updates, and mask pair zero. Selection scores average four noise samples, while FM8 displays the first. Section 5.6 describes the selection rules.

6 Conclusion

Our study shows that increasing the sampling budget does not consistently improve image inpainting quality, and that pixel fidelity and perceptual similarity can favor different inference budgets. In the development experiments, two function evaluations achieve the highest PSNR among the tested settings, whereas eight achieve the lowest LPIPS. The development-selected eight-evaluation FM model improves both mean metrics over pixel and perceptual regression on the test set, at approximately 7.4 times the single-image inference latency of perceptual regression. These findings support selecting sampling budgets according to the target quality criterion and computational cost. The current study examines 64×64 Oxford-IIIT Pet images with Euler sampling. Future work can test whether the observed budget preferences and relationship between quality and computational cost extend to higher resolutions, other image domains, and alternative integration schemes.

Author contributions

Yongji Chen led the study design and manuscript writing. He formulated the research questions, designed the controlled comparisons and evaluation protocol, and worked with Kaichun You to interpret the results. He organized the manuscript and integrated the methods, findings, and limitations into the final narrative.

Kaichun You led model implementation, training, and experimental analysis. His responsibilities included data preparation and preprocessing, matched-mask construction, implementation of the regression and flow-matching models, and the training and evaluation pipelines. He conducted the experiments, analyzed training dynamics and reconstruction metrics, and documented the implementation settings for reproducibility.

Linwei Zhang contributed to the literature review by organizing relevant work on flow matching, image inpainting, and perceptual evaluation. He helped develop the related-work discussion, clarify the relationship between prior studies and the project, and check the supporting references.

Zijun Tang assisted with visualization and manuscript revision. He helped organize the quantitative plots and qualitative examples, improve figure captions and presentation, and review the report for clarity and consistency.

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